MODULES OVER A FINITE SIMPLE ALGEBRA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for all six assertions; each is attributed to the named structure theorem, the named equivalence or an explicit description of the module, and no step is left to the reader.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Editorial note. The shared gate prints 'occupancy: published-overlap: brauer.tex 483-508' for this candidate while still returning RESULT=OK. That is the known cross-coordinate false overlap of LANE_BRIEF 10.5: the public item 062 carries upstream RAW line numbers (483-508) while this card carries pinned coordinates, and for brauer the pinned file has an inline preamble of 221 lines, so the raw interval corresponds to pinned lines about 700-725. The true public proposition there is theorem-skolem-noether (Stacks tag 074Q), which this lane excluded by both the gate and the tag intersection; the tag of this card is not the public tag, so this card is not a duplicate.

Allowed standard prerequisites (editorial).

- Wedderburn's theorem. Same chapter, Theorem theorem-wedderburn. Supplies the description of a finite simple algebra as a matrix algebra over a finite skew field extension of the ground field, which reduces everything to modules over a skew field.
- Equivalence of module categories for a matrix algebra. Same chapter, Lemma lemma-matrix-algebras. Supplies the equivalence between modules over the matrix algebra and modules over the skew field, under which the statements about simple modules, direct sums and dimensions are transported.

Source reference. Original theorem 0747: sources/stacks-brauer/pinned-original-source.tex, source lines 358–358. Navigation pointer invoked at proof line 519;

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the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 074D: sources/stacks-brauer/pinned-original-source.tex, source lines 477–477. Navigation pointer invoked at proof line 522; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let A be a finite simple k-algebra.*

(1) *There exists exactly one simple A-module M up to isomorphism.*
(2) *Any finite A-module is a direct sum of copies of a simple module.*
(3) *Two finite A-modules are isomorphic if and only if they have the same dimension over k.*
(4) *If $A = Mat(n \times n, K)$ with K a finite skew field extension of k, then $M = K^{\oplus n}$ is a simple A-module and $End_A(M) = K^{op}$.*
(5) *If M is a simple A-module, then $L = End_A(M)$ is a skew field finite over k acting on the left on M, we have $A = End_L(M)$, and the centers of A and L agree. Also $[A : k][L : k] = \dim_k(M)^2$.*
(6) *For a finite A-module N the algebra $B = End_A(N)$ is a matrix algebra over the skew field L of (5). Moreover $End_B(N) = A$.*

Proof. By Theorem 0747 we can write $A = \mathrm{Mat}(n \times n, K)$ for some finite skew field extension K of k. By Lemma 074D the category of modules over A is equivalent to the category of modules over K. Thus (1), (2), and (3) hold because every module over K is free. Part (4) holds because the equivalence transforms the K-module K to $M = K^{\oplus n}$. Using $M = K^{\oplus n}$ in (5) we see that $L = K^{op}$. The statement about the center of $L = K^{op}$ follows from Lemma 074D. The statement about $\mathrm{End}_L(M)$ follows from the explicit form of M. The formula of dimensions is clear. Part (6) follows as N is isomorphic to a direct sum of copies of a simple module. □